

GRAND JUNCTION, CO, USA

WMO: 724760

Lat: 39.1200N

Lon: 108.5250W

Elev: 4833

StdP: 12.30

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 23066

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.9	(c) 10.9	(d) -3.0	(e) 5.6	(f) 16.7	(g) 1.1	(h) 6.9	(i) 23.5	(j) 19.8	(k) 38.7	(l) 17.6	(m) 34.5	(n) 4.1	(o) 80	(p) 0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.9	97.8	61.2	95.5	60.3	93.1	59.8	65.2	85.3	64.1	84.3	63.1	83.3	10.2	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.9	95.6	68.0	58.8	88.7	68.6	56.5	81.4	69.9	32.6	84.4	31.7	83.9	30.9	83.0	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 23.7	(b) 19.6	(c) 17.2	(e) 0.2	(f) 101.7	(g) 7.0	(h) 1.8	(i) -4.8	(j) 103.0	(k) -8.8	(l) 104.1	(m) -12.8	(n) 105.1	(o) -17.8	(p) 106.4		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.6	28.6	35.6	45.2	52.0	61.9	73.4	79.7	76.4	67.3	53.3	40.2	28.9
	DBStd	18.94	8.25	7.54	7.53	7.80	8.00	6.75	4.44	4.18	7.05	7.77	7.79	8.52
	HDD50	2347	662	405	187	71	8	0	0	0	1	55	304	654
	HDD65	5416	1127	824	615	392	154	16	0	0	56	369	745	1118
	CDD50	3672	0	1	37	131	377	703	919	818	520	157	9	0
	CDD65	1266	0	0	0	2	58	269	455	353	124	5	0	0
	CDH74	15879	0	0	6	79	926	3671	5713	3950	1443	91	0	0
	CDH80	7880	0	0	0	8	289	1899	3185	1954	534	11	0	0
	WSAvg	7.5	5.1	6.3	7.8	9.0	8.9	9.4	9.0	8.5	8.2	7.1	5.8	5.0
	PrecAvg	8.9	0.6	0.5	0.8	0.9	0.9	0.4	0.6	0.9	1.0	1.0	0.7	0.6
Precipitation	PrecMax	14.8	1.7	1.3	2.0	2.3	2.0	2.1	1.9	3.1	3.1	3.5	2.0	2.0
	PrecMin	4.5	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0
	PrecStd	2.3	0.4	0.4	0.6	0.6	0.6	0.5	0.5	0.6	0.8	0.7	0.4	0.4
	PrecStd	2.3	0.4	0.4	0.6	0.6	0.6	0.5	0.5	0.6	0.8	0.7	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.6	61.5	74.1	80.3	91.7	99.6	101.4	98.1	93.2	81.2	66.4	53.9
		MCWB	40.4	44.5	48.5	51.6	57.2	60.0	62.5	61.6	60.4	54.3	47.4	41.9
	2%	DB	47.5	56.0	69.0	76.1	86.5	96.1	98.9	95.5	89.3	76.3	62.4	48.3
		MCWB	38.0	41.4	46.4	49.4	54.9	58.5	62.0	60.7	58.8	52.4	45.0	39.0
	5%	DB	43.8	51.7	64.7	72.8	82.9	93.5	96.6	93.0	86.2	72.1	58.3	44.5
		MCWB	36.1	39.7	44.6	48.3	53.6	57.7	61.4	60.2	57.2	50.8	43.4	36.4
	10%	DB	40.3	47.7	60.6	68.8	79.5	90.7	94.3	90.3	82.9	68.0	53.9	40.7
		MCWB	34.2	37.7	42.8	46.9	52.5	57.0	60.9	60.3	56.3	49.0	41.7	34.3
	0.4%	WB	42.2	45.9	50.0	53.2	60.2	63.1	66.9	66.4	63.7	57.8	49.3	43.7
		MCDB	47.7	57.6	69.9	75.7	85.0	89.5	86.7	82.6	81.9	70.6	63.5	51.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.3	42.9	47.6	51.0	57.0	61.3	65.5	65.1	61.8	55.0	46.4	40.5
		MCDB	45.2	52.7	66.2	71.4	79.4	86.9	87.0	83.1	78.7	69.9	58.8	47.2
	5%	WB	37.1	41.0	45.6	49.4	55.1	59.9	64.4	64.1	60.4	53.0	44.4	37.3
		MCDB	42.6	49.6	61.5	68.7	77.2	86.2	86.1	81.9	76.6	66.8	55.8	43.0
	10%	WB	34.8	38.9	43.8	47.9	53.5	58.6	63.6	63.0	59.0	51.0	42.3	34.8
		MCDB	39.3	46.5	58.6	66.2	75.8	84.7	85.3	81.0	75.1	64.8	52.5	39.9
Mean Daily Temperature Range	5% DB	MDBR	18.4	19.8	24.3	24.9	26.9	29.4	27.9	26.6	26.4	24.7	22.1	18.1
		MCDBR	21.4	25.2	30.0	30.7	31.5	32.3	31.0	30.4	30.3	29.6	27.5	21.4
		MCWBR	13.3	13.7	14.3	13.5	12.6	12.3	11.1	11.0	11.8	13.7	14.9	13.4
	5% WB	MCDBR	18.1	22.0	27.5	27.8	28.9	29.1	26.0	23.9	23.3	24.1	24.4	18.8
		MCWBR	11.8	12.5	13.5	12.6	11.5	11.5	8.9	8.3	9.3	11.5	13.5	12.3
Clear-Sky Solar Irradiance	taub		0.255	0.260	0.296	0.315	0.327	0.323	0.368	0.372	0.321	0.297	0.267	0.256
	taud		2.561	2.569	2.466	2.438	2.457	2.469	2.389	2.377	2.514	2.547	2.573	2.548
	Ebn at Noon		299	312	307	305	302	301	287	283	294	292	292	290
	Edh at Noon		23	26	32	35	35	35	37	37	30	26	22	21
All-Sky Solar Radiation	RadAvg		817	1075	1511	1896	2179	2516	2256	1989	1732	1268	902	693
	RadStd		52	81	113	115	146	147	114	90	88	100	65	55

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	-2.66	-1.64	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page